



Universidad
Tecnológica
del Perú

PRIMERA UNIDAD DE APRENDIZAJE 2: Bases de Datos Distribuidas

CURSO: BASE DE DATOS II

SESIÓN N° 9: ORACLE APEX

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**DOCENTES DE UTP
SISTEMAS**

Registro Oracle Apex

<https://www.oracle.com/apex/>

Registro con correo
universitario



First name *

Last name *

Email *

Workspace name *

The workspace name uniquely identifies your development environment and is used when signing in.

Country *

Are you new to Oracle APEX? *

☐ Yes ☐ No

Do you plan to use APEX for education or training? *

☐ Yes ☐ No

☐ I agree to the terms of the [Oracle APEX Service Agreement](#)

By filling out and submitting this form, you understand and agree that the use of Oracle's website is subject to the [Oracle.com Terms of Use](#). Additional details regarding Oracle's collection and use of your personal information, including information about access, retention, rectification, deletion, security, cross-border transfers, and other topics, is available in the [Oracle Privacy Policy](#).

Request Workspace

Registro Oracle Apex

<https://www.oracle.com/apex/>



Welcome
Sign in to your workspace to continue

 Workspace

 Username

 Password 

☐ Remember Workspace and Username 

Sign In

ORACLE APEX: Creación de DASHBOARD

1. SEDE

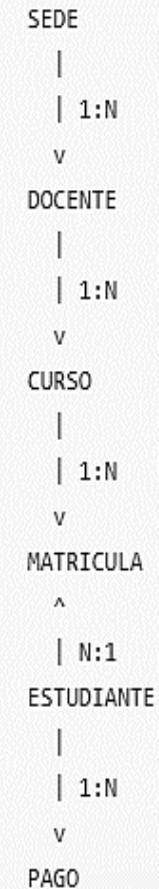
Representa las sedes de la institución.

Campo	Tipo
id_sede	NUMBER
nombre	VARCHAR2(100)
ciudad	VARCHAR2(100)

2. DOCENTE

Campo	Tipo
id_docente	NUMBER
nombres	VARCHAR2(100)
especialidad	VARCHAR2(100)
id_sede	NUMBER

Relaciones generales



Relación

- Una sede tiene muchos docentes.
- 1:N → SEDE → DOCENTE

3. CURSO

Campo	Tipo
id_curso	NUMBER
nombre	VARCHAR2(100)
creditos	NUMBER
id_docente	NUMBER

4. ESTUDIANTE

Campo	Tipo
id_estudiante	NUMBER
nombres	VARCHAR2(100)
correo	VARCHAR2(100)
ciclo	NUMBER

Relación

- Un docente dicta muchos cursos.
- 1:N → DOCENTE → CURSO

5. MATRICULA

Campo	Tipo
id_matricula	NUMBER
fecha_matricula	DATE
id_estudiante	NUMBER
id_curso	NUMBER

Relación

- Una matrícula puede tener muchos pagos.
- 1:N → MATRICULA → PAGO

Relación

- Un estudiante puede tener muchas matrículas.
- Un curso puede tener muchas matrículas.

6. PAGO

Campo	Tipo
id_pago	NUMBER
monto	NUMBER(10,2)
fecha_pago	DATE
id_matricula	NUMBER

ORACLE APEX: Creación de DASHBOARD





App Builder
Create and manage applications.



SQL Workshop
Develop and manage database objects.



Data Reporter
Create reporting applications based on datasets.



Gallery
Install and explore sample applications.

Top Apps



DashboardEjemplo2

45



DashboardN°1BD

28

Top Users



C31052

208

Summary

8 Applications

15 Tables

1 Developers

0 Reporting Applications

Workspace Message 



Object Browser

Browse and manage database objects such as tables, views, and packages.



SQL Commands

Execute ad hoc SQL statements against workspace schemas.



SQL Scripts

Create, organize, and run stored SQL scripts.



Utilities

Access data loading and database utilities.



RESTful Services

Configure and manage RESTful services and endpoints.

```
1 CREATE TABLE sede (  
2     id_sede NUMBER PRIMARY KEY,  
3     nombre VARCHAR2(100),  
4     ciudad VARCHAR2(100)  
5 );
```

```
7 CREATE TABLE docente (  
8     id_docente NUMBER PRIMARY KEY,  
9     nombres VARCHAR2(100),  
10    especialidad VARCHAR2(100),  
11    id_sede NUMBER,  
12    CONSTRAINT fk_docente_sede  
13    FOREIGN KEY (id_sede)  
14    REFERENCES sede(id_sede)  
15 );
```

```
17 CREATE TABLE curso (  
18     id_curso NUMBER PRIMARY KEY,  
19     nombre VARCHAR2(100),  
20     creditos NUMBER,  
21     id_docente NUMBER,  
22     CONSTRAINT fk_curso_docente  
23     FOREIGN KEY (id_docente)  
24     REFERENCES docente(id_docente)  
25 );
```

```
27 CREATE TABLE estudiante (  
28     id_estudiante NUMBER PRIMARY KEY,  
29     nombres VARCHAR2(100),  
30     correo VARCHAR2(100),  
31     ciclo NUMBER  
32 );
```

```
34 CREATE TABLE matricula (  
35     id_matricula NUMBER PRIMARY KEY,  
36     fecha_matricula DATE,  
37     id_estudiante NUMBER,  
38     id_curso NUMBER,  
39     CONSTRAINT fk_matricula_estudiante  
40     FOREIGN KEY (id_estudiante)  
41     REFERENCES estudiante(id_estudiante),  
42  
43     CONSTRAINT fk_matricula_curso  
44     FOREIGN KEY (id_curso)  
45     REFERENCES curso(id_curso)  
46 );
```

```
48 CREATE TABLE pago (  
49     id_pago NUMBER PRIMARY KEY,  
50     monto NUMBER(10,2),  
51     fecha_pago DATE,  
52     id_matricula NUMBER,  
53     CONSTRAINT fk_pago_matricula  
54     FOREIGN KEY (id_matricula)  
55     REFERENCES matricula(id_matricula)  
56 );
```

```
59
60 INSERT INTO sede VALUES (1,'Sede Lima','Lima');
61 INSERT INTO sede VALUES (2,'Sede Norte','Trujillo');
62
63 INSERT INTO docente VALUES (1,'Carlos Perez','Java',1);
64 INSERT INTO docente VALUES (2,'Ana Torres','Base de Datos',2);
65
66 INSERT INTO curso VALUES (1,'Java Web',4,1);
67 INSERT INTO curso VALUES (2,'Oracle SQL',3,2);
68
69 INSERT INTO estudiante VALUES (1,'Luis Mendoza','luis@gmail.com',3);
70 INSERT INTO estudiante VALUES (2,'Maria Lopez','maria@gmail.com',5);
71
72 INSERT INTO matricula VALUES (1,SYSDATE,1,1);
73 INSERT INTO matricula VALUES (2,SYSDATE,2,2);
74
75 INSERT INTO pago VALUES (1,450,SYSDATE,1);
76 INSERT INTO pago VALUES (2,500,SYSDATE,2);
```

```
81  SELECT
82      c.nombre AS curso,
83      d.nombres AS docente
84  FROM curso c
85  INNER JOIN docente d
86  ON c.id_docente = d.id_docente;
87
```

```
89  SELECT
90      e.nombres AS estudiante,
91      c.nombre AS curso,
92      m.fecha_matricula
93  FROM matricula m
94  INNER JOIN estudiante e
95  ON m.id_estudiante = e.id_estudiante
96  INNER JOIN curso c
97  ON m.id_curso = c.id_curso;
98
```

```
100 SELECT
101     e.nombres,
102     SUM(p.monto) AS total_pagado
103 FROM pago p
104 INNER JOIN matricula m
105 ON p.id_matricula = m.id_matricula
106 INNER JOIN estudiante e
107 ON m.id_estudiante = e.id_estudiante
108 GROUP BY e.nombres;
```

```
110  SELECT
111      |      c.nombre AS curso,
112      |      COUNT(m.id_estudiante) AS cantidad_estudiantes
113  FROM curso c
114  LEFT JOIN matricula m
115  ON c.id_curso = m.id_curso
116  GROUP BY c.nombre;
```

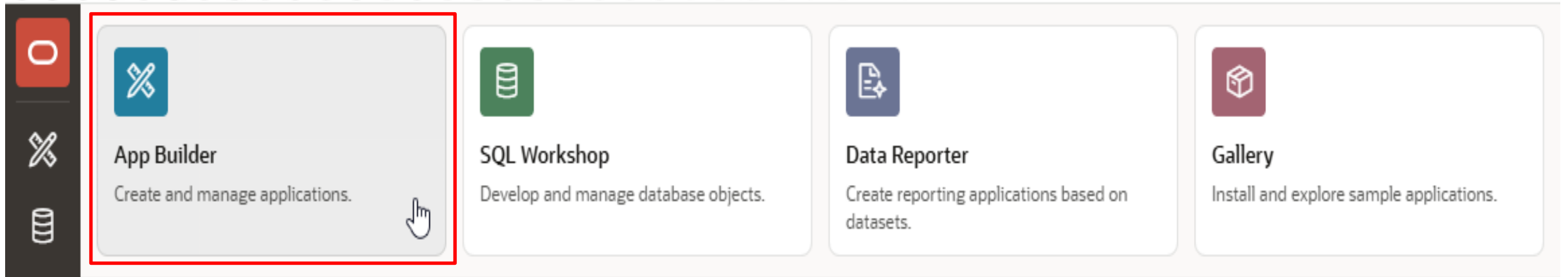
```
119  SELECT
120      |      c.nombre,
121      |      COUNT(m.id_estudiante) cantidad
122  FROM curso c
123  LEFT JOIN matricula m
124  ON c.id_curso = m.id_curso
125  GROUP BY c.nombre;
```

```
127  SELECT
128      |      e.nombres,
129      |      SUM(p.monto) total
130  FROM pago p
131  INNER JOIN matricula m
132  ON p.id_matricula = m.id_matricula
133  INNER JOIN estudiante e
134  ON m.id_estudiante = e.id_estudiante
135  GROUP BY e.nombres;
```

```
137 SELECT
138     e.nombres estudiante,
139     c.nombre curso,
140     d.nombres docente,
141     p.monto
142 FROM estudiante e
143 INNER JOIN matricula m
144 ON e.id_estudiante = m.id_estudiante
145 INNER JOIN curso c
146 ON m.id_curso = c.id_curso
147 INNER JOIN docente d
148 ON c.id_docente = d.id_docente
149 INNER JOIN pago p
150 ON m.id_matricula = p.id_matricula;
```

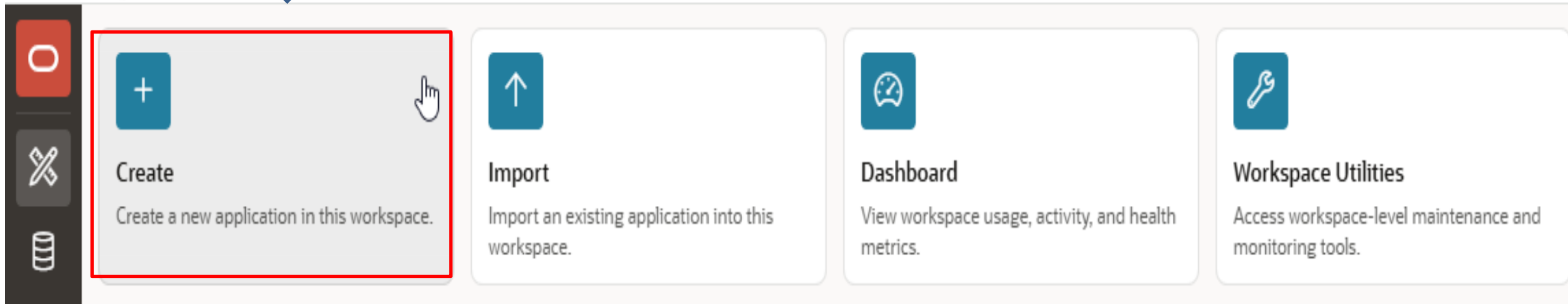
```
152 SELECT
153     c.nombre AS curso,
154     d.nombres AS docente,
155     SUM(p.monto) AS monto
156 FROM estudiante e
157 INNER JOIN matricula m
158 ON e.id_estudiante = m.id_estudiante
159 INNER JOIN curso c
160 ON m.id_curso = c.id_curso
161 INNER JOIN docente d
162 ON c.id_docente = d.id_docente
163 INNER JOIN pago p
164 ON m.id_matricula = p.id_matricula
165 GROUP BY c.nombre, d.nombres
166 ORDER BY c.nombre
```

```
169  SELECT
170      |      e.nombres estudiante,
171      |      SUM(p.monto) monto
172  FROM estudiante e
173  INNER JOIN matricula m
174  ON e.id_estudiante = m.id_estudiante
175  INNER JOIN pago p
176  ON m.id_matricula = p.id_matricula
177  GROUP BY e.nombres
```



This screenshot shows the Oracle APEX workspace menu. On the left is a vertical sidebar with three icons: a red circle with a white 'O', a crossed wrench and screwdriver, and a database cylinder. The main area contains four tiles. The first tile, 'App Builder', is highlighted with a red border and has a mouse cursor icon over it. The other tiles are 'SQL Workshop', 'Data Reporter', and 'Gallery'.

- App Builder**
Create and manage applications.
- SQL Workshop**
Develop and manage database objects.
- Data Reporter**
Create reporting applications based on datasets.
- Gallery**
Install and explore sample applications.



This screenshot shows the Oracle APEX workspace menu after clicking 'App Builder'. The sidebar is the same. The main area now shows four different tiles: 'Create', 'Import', 'Dashboard', and 'Workspace Utilities'. The 'Create' tile is highlighted with a red border and has a mouse cursor icon over it.

- Create**
Create a new application in this workspace.
- Import**
Import an existing application into this workspace.
- Dashboard**
View workspace usage, activity, and health metrics.
- Workspace Utilities**
Access workspace-level maintenance and monitoring tools.

Create an Application

Name

ID

82859

Create Application



Use Create App Wizard

Create a new application using the full Create Application wizard with advanced options.



Create App From a File

Upload a CSV, XLSX, XML or JSON file, or copy and paste data, then create your application.



Create Fusion Integration

Create a starter application with pre-configured integrations to an Oracle Fusion Applications instance.

Create an Application



Name

DashboardEjemploDB

Appearance

Iris, Side Menu



Pages ?



Add Page



Home

Blank

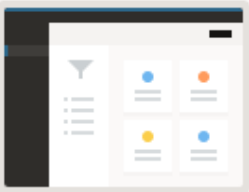
Edit



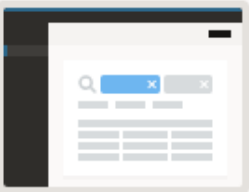
Add Page



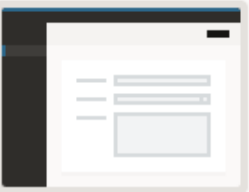
Dashboard



Faceted Search



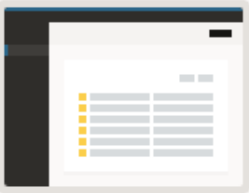
Smart Filters



Form



Interactive Grid



Interactive Report



Map

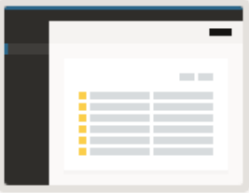


Master Detail

▼ Additional Pages



Classic Report



Multiple Reports



Timeline



Wizard

Add Report Page

Page Name

Table or View SQL Query Interactive Report Classic Report

Table or View
- Select Table or View -

Add Report Page

Page Name
Grafico N°1

Table or View SQL Query Interactive Report Classic Report

```
1 SELECT
2   e.nombres estudiante,
3   SUM(p.monto) monto
4 FROM estudiante e
5 INNER JOIN matricula m
6 ON e.id_estudiante = m.id_estudiante
7 INNER JOIN pago p
8 ON m.id_matricula = p.id_matricula
9 GROUP BY e.nombres
```

Create an Application



Name

DashboardEjemploDB

Appearance

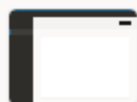
Iris, Side Menu



Pages [?]



Add Page



Home

Blank

Edit



Grafico N°1

Classic Report (Custom Query)

Edit



Features [?] Check All

Cancel

Create Application

Creating classic report page






Application 82859





DashboardEjemploDB
Edit Application Definition



Run Application
Run this application in a new browser window.



Supporting Objects
Manage installation scripts and supporting database objects.



Shared Components
Define and manage shared UI components, data sources, and application logic.



Utilities
Access diagnostic, monitoring, and maintenance utilities.



Export / Import
Export or import this application between environments.




Actions


Create Page

Page ↑≡	Name	Alias	Updated	Updated By	Type	Group	Actions
0	Global Page	-	53 seconds ago	c31052@utp.edu.pe	Global Page	Unassigned	 
1	Home	home	49 seconds ago	c31052@utp.edu.pe	Home	Unassigned	 
2	Grafico N°1	grafico-n-1	49 seconds ago	c31052@utp.edu.pe	Report	Unassigned	 
9999	Login Page	login	53 seconds ago	c31052@utp.edu.pe	Login	Unassigned	 

←
✉
Application 82859 \ Page 2: Grafico N°1

📄
2
⬆
⬇
🔒
↺
↻
+
⚙
🏠
Save
▶

📁
⚡
↺
🏠

📄
Page 2: Grafico N°1
☰

> Pre-Rendering
> Components
> Breadcrumb Bar
> 🔗 Breadcrumb
> Body
> 📊 Grafico N°1
> Columns
> Post-Rendering

Layout
Page Search
Help

🔍
⊕
↗
☰

📊
Grafico N°1

COPY

EDIT

PREVIOUS

NEXT

SORT ORDER

REGION BODY

REGION CONTENT

Regions

Items

Buttons

◇ Actions
👤 Avatar
🏷 Badge
🔗 Breadcrumb
📅 Calendar

📄 Cards
📊 Chart
📄 Classic Report
💬 Comments
✓ Content Row

📄 Dynamic Content
📄 Faceted Search
📄 Flexbox
📄 Form
❓ Help Text

Region
Attributes

🔍
Filter
↶

> Identification
Name
Grafico N°1
Title
Type
Classic Repc
⋮

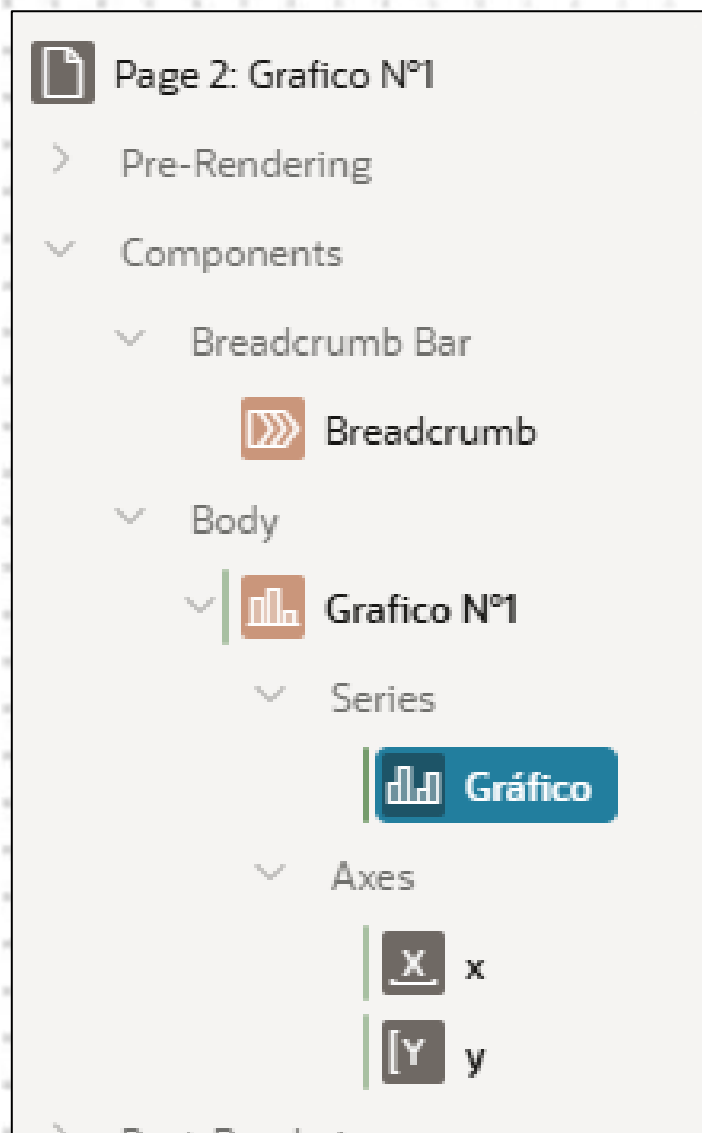
> Source
Location
Local Database
Type
SQL Query
SQL Query
🔗

```

SELECT
  e.nombres estudiante,
  SUM(p.monto) monto
FROM estudiante e
INNER JOIN matricula m
ON e.id_estudiante = m.id_estudiante
INNER JOIN pago p
ON m.id_matricula = p.id_matricula
GROUP BY e.nombres

```

Page Items to Submit
⋮



Region Attributes

Filter

▼ Identification

Name Grafico N°1

Title

Type Classic Repc

▼ Source

Location

Type

SQL Query

SELECT
e.
SU
FROM e
INNER
ON e.i
INNER
ON m.i
GROUP

Page Item

Submit

Optimiz

Built-in Components

- Breadcrumb
- Calendar
- Cards
- Chart**
- Classic Report
- Dynamic Content
- Faceted Search
- Form
- Help Text
- Interactive Grid
- Interactive Report
- List
- Map
- Region Display Selector

Page 2: Grafico N°1

> Pre-Rendering

Components

Breadcrumb Bar

Breadcrumb

Body

Grafico N°1

Series

Gráfico

Axes

x

y

Identification

Name Gráfico

Execution

Sequence 10

Source

Location Local Database

Type SQL Query

SQL Query

```
SELECT
    e.nombres estudiante,
    SUM(p.monto) monto
FROM estudiante e
INNER JOIN matricula m
ON e.id_estudiante = m.id_estudiante
INNER JOIN pago p
ON m.id_matricula = p.id_matricula
GROUP BY e.nombres
```

Column Mapping

Series Name - Select -

Label ESTUDIANTE

Value MONTO

Z - Select -

Custom Tooltip - Select -

Legend Tooltip - Select -

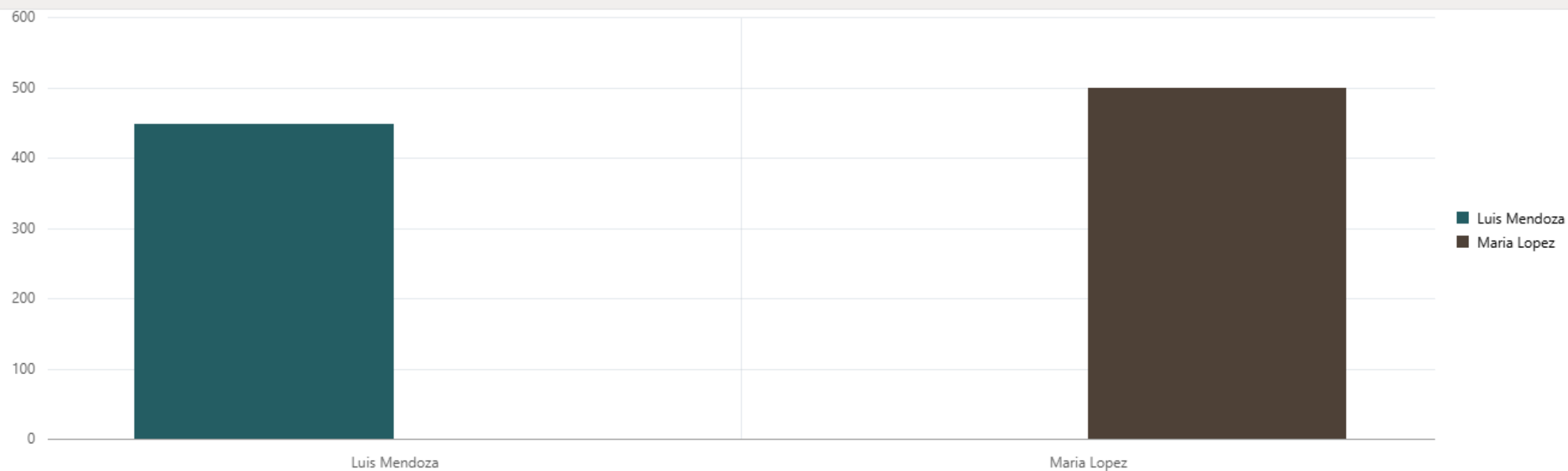
Label Tooltip - Select -

Custom Column - Select -

Save

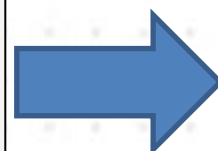


Grafico N°1



Page 2: Grafico N°1

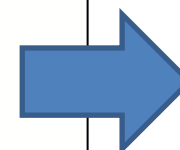
- Pre-Rendering
- Components
 - Breadcrumb Bar
 - Breadcrumb
 - Body
 - Grafico N°1
 - Series
 - Gráfico
 - Axes
 - x
 - y
 - Gráfico N°2
 - Series
 - Gráficos
 - Axes
 - x
 - y



Code Editor - SQL Query

```

1  SELECT
2      c.nombre AS curso,
3      d.nombres AS docente,
4      SUM(p.monto) AS monto
5  FROM estudiante e
6  INNER JOIN matricula m
7  ON e.id_estudiante = m.id_estudiante
8  INNER JOIN curso c
9  ON m.id_curso = c.id_curso
10 INNER JOIN docente d
11 ON c.id_docente = d.id_docente
12 INNER JOIN pago p
13 ON m.id_matricula = p.id_matricula
14 GROUP BY c.nombre, d.nombres
15 ORDER BY c.nombre
  
```



Column Mapping

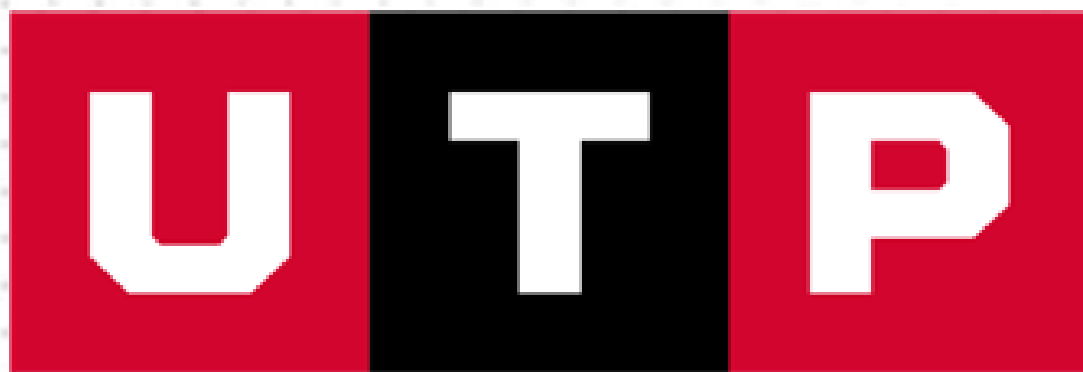
Series Name	DOCENTE
Label	CURSO
Value	MONTO
Z	- Select -
Custom Tooltip	- Select -
Legend Tooltip	DOCENTE
Label Tooltip	MONTO
Custom Column	- Select -

Save



Gráfico N°2





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Tecnológica
del Perú**

